Accessibility Matrix

Terms and Conditions

Purpose: protect the software's creators and set the limits of their liability.

1. Terms of use of the service

Permitted

- Each role (Owner, Maintenance Company, Inspection Company, CMA Staff) uses only the features and elevators linked to its own account.
- The Maintenance Company registers elevators and complaints; the Inspection Company schedules inspections and submits results and reports.

Prohibited

- Entering false data or uploading altered reports.
- Sharing login credentials or accessing other users' data.
- Trying to get around access control or overloading the system.

Why: the platform feeds the municipal elevator register and decisions about people's safety. Each user is responsible for the accuracy of what they submit, not the creators. This also justifies suspending accounts in case of misuse.

2. Liability disclaimers

- The service is provided "as is", the creators are not liable for losses caused by failures, downtime or data loss.
- **Notifications:** reminders (e.g. the notice 60 days before the deadline) and alerts are only an aid. They do not replace the legal obligations of owners and maintenance companies (Decree-Law no. 320/2002) or the Municipality's official notices.
- **User content:** the creators do not check the technical content of reports or of data entered by users.

Why: this is the most critical point. If a reminder fails and an inspection becomes overdue, or an unsafe elevator stays in service, the responsibility cannot fall on the people who built the software.

3. Refund / cancellation

- **Refunds: not applicable.** The platform is free for its users. Inspection fees are paid to the Municipality outside the platform, under the municipal regulation.

- **Account cancellation:** users may ask for their account to be closed. Elevator and inspection records stay in the municipal register for as long as the law requires. The CMA may suspend accounts that breach these terms.

Why: it makes clear that the creators are not involved in payments. It also explains why an elevator's history does not disappear when an account is closed, which keeps the register intact.

4. Governing law and jurisdiction

- Portuguese law applies, in particular Decree-Law no. 320/2002 (maintenance and inspection of elevators) and the GDPR (with Law no. 58/2019).
- The courts of the district of Aveiro have jurisdiction.

Why: the service is provided in a Portuguese municipality to Portuguese users. Naming the court avoids uncertainty about where, and under which law, a dispute is settled.

5. Changes to the terms

- The terms may change, for example because of changes in the law or new features.
- Users are notified (by email or a notice on the platform) at least 15 days before changes take effect.
- Significant changes must be accepted again at the next login. The accepted version and the date are recorded.
- Continuing to use the platform after that counts as acceptance.

Why: the project will evolve (MVP, then extended features, then deployment at the CMA). The terms need to keep up with that evolution without losing the validity of users' acceptance.

Legal Requirements

Elevator inspections in Portugal are governed by Decreto-lei nº320/2002, which makes municipalities responsible for inspecting lifts in their area; new buildings also need a use permit under the RJUE (Decreto-lei nº555/99). Elevate supports the CMA in applying these rules, but it does not replace the inspection company or the CMA's decisions. The table lists the requirements that shape the platform and how it meets each one.

Legal requirement	**How Elevate complies**

Lifts must be inspected periodically (Decreto-lei nº320/2002, art. 8); the CMA applies a 2-year period	Each elevator keeps the validity date of its last certificate, and its status changes to "due soon" or "overdue" on its own. The 2-year rule lives in one configurable module (see Technical Debt)
The inspection must be requested before the certificate expires (Anexo V, n.º 1); the CMA asks for the request 2 months in advance	Owners and maintenance companies are notified 60 days before the deadline (FR4). The owner submits the request at the CMA desk, where staff enter it in AIRC's MyNet (MyDoc WIN), or online; Elevate receives and processes it without changing that system (NFR5)
The request comes with proof that the inspection fee has been paid (Anexo V, n.º 1.1)	The payment reference is generated and paid in MyNet, outside Elevate; the platform does not handle payments
Deadlines count from entry into service for new lifts and from the last periodic inspection for the others (Anexo V, n.º 3)	The registration records the entry-into-service date, and later deadlines come from the last certificate
The inspection takes place within 60 days of the request (Anexo V, n.º 1.2)	The inspection company schedules each request in the platform and every party is notified (FR5); the CMA sees pending and overdue requests on its dashboard
A lift that passes gets a certificate stating when the next inspection is due, and the owner, the CMA and the maintenance company are informed (Anexo V, n.os 4 e 5)	The inspection company submits the result, the certificate validity date and the report or certificate in PDF (FR6); all three parties see it in the elevator history
A lift that fails has up to 30 days to be corrected, followed by a re-inspection (Anexo V, n.os 6 e 6.1)	The inspection company records the non-conformities and the correction deadline; the maintenance company marks the corrections as done, the owner requests the re-inspection, and the CMA follows each case (FR7)
Every lift must have a maintenance contract with a maintenance company (art. 4)	Each elevator is registered by its maintenance company (FR2), which can attach the contract, so every lift in the register is linked to one
The municipality oversees the lifts in its area (art. 7)	The CMA approves a registration only if it matches a prior notice, gives it an internal reference (FR3) and alerts owners of lifts that are still unregistered
New buildings need a use permit from the municipality (RJUE, Decreto-lei n 555/99)	When a use permit is issued, Urban Planning Sector staff log a notice with the building address and number of elevators, which alerts the elevator division to follow up the registration (FR1)
Users may report a faulty lift to the municipality, which may order an extraordinary inspection (art. 8, n.º 6)	The maintenance company registers complaints, which are kept in the elevator history and notified to the owner and the CMA
Public administration online services must let citizens authenticate with the Citizen Card or Digital Mobile Key (Lei n.º 37/2014, Lei n.º 7/2007, eIDAS Regulation)	Owners log in with Autenticação.gov through Authentik. The CMA registers Elevate with ARTE, IP (formerly AMA), and Elevate only receives the attributes the owner authorizes (name and tax number)

Personal Data Protection (GDPR)

Elevate stores personal data of owners, condominiums and the staff of the CMA and of the maintenance and inspection companies, so it must comply with Regulation (EU) 2016/679 (GDPR) and Law No. 58/2019, which implements it in Portugal. The Municipality of Aveiro is the data controller, and its Data Protection Officer can be reached at protecao-de-dados@cm-aveiro.pt. The data is processed because the CMA is legally required to oversee the lifts in its area (RGPD art. 6(1) (c) e (e), Decreto-lei nº320/2002), so consent is not needed.

Personal data and purpose

Data	Whose	Purpose
Name	All users; owners and condominiums	Identify who performs each action and who is responsible for each elevator
Email	All users	Automatic notices (deadlines, scheduling, results) and account identification
Phone (optional)	Owners, maintenance and inspection companies	Contact in urgent situations or to schedule visits
Tax number (NIF)	Owners, condominiums and companies	Identify the person responsible for the elevator, as the CMA already does today
Address	Only the elevator's address	Locate the elevator and match it to the Urban Planning notice. The owner's home address is not stored
Password	Maintenance companies, inspection companies and CMA staff	Login. Managed by Authentik and stored only as a hash. Owners have no password in Elevate: they log in with Autenticação.gov

Data minimization: development and staging use test data only, and real records are only loaded on the CMA's servers. The dashboard shows totals only, with no personal data. The phone number and the contact of the resident who made a complaint are optional, and the technical characteristics of the elevator and the owner's home address are not collected.

Retention and right to erasure

Data	Retention period	Reason
User accounts (email, phone, password)	Deleted automatically 12 months after deactivation	Allows the account to be reactivated or pending processes to be closed
Access logs	Deleted automatically after 12 months	Security and audit
Elevator history and inspection records	Kept permanently	Part of the CMA's archive; the GDPR allows longer retention for archiving in the public interest (art. 5(1)(e) e art.

		89)

A data subject can request erasure through Elevate or the CMA's Data Protection Officer, who replies within 30 days. The account is deactivated and the data that is no longer needed is deleted automatically at the end of the retention period, or immediately if there are no pending processes. Inspection records are not deleted, as they are part of the CMA's archive and of a legal obligation (RGPD art. 17(3)): the user's name is replaced by "Removed user" and the tax number of the person responsible for the elevator is kept.

Sharing with third parties

Party	**Data it can access**	**Purpose**
Maintenance company	Only its own elevators: address, owner's name and contact, inspection history	Follow its elevators' inspections and record corrections after a failed inspection
Inspection company	Only the inspections assigned to it: elevator address and contact for scheduling	Schedule and carry out the inspection
MyDoc WIN (AIRC)	The CMA's document management system, where inspection requests are registered. Elevate receives the request data (elevator, owner, scope) and does not write to it	Bring the inspection requests made at the CMA into Elevate
Autenticação.gov (ARTE, IP, formerly AMA)	Authenticates owners; Elevate receives only the attributes the owner authorizes (name and tax number)	Login with Citizen Card or Digital Mobile Key

Authentik runs on the CMA's own infrastructure, so login data is not shared with an external provider. Data is not shared with any other party, is not used for advertising and is not transferred outside the European Union.

Security measures

Measure	**How Elevate applies it**
Access control	Role-based and enforced on the server, not only in the interface: each profile only sees and does what it needs
Authentication	Authentik manages accounts, roles and passwords (stored as hashes). Owners log in with Autenticação.gov through Authentik; the other profiles use email and password
Encryption	In transit (HTTPS) and at rest, using only open-source features

Audit log	Every change to inspection records is logged
Input validation	User inputs and uploaded files are validated
Backups	Automatic backups of the database and documents, stored outside the server that hosts the platform, with the restore process tested regularly
Data breaches	Reported to the supervisory authority (CNPD) within 72 hours, and to the data subjects when there is a high risk to them (RGPD art. 33 e 34)

To confirm with the CMA: that inspection records are kept permanently in the municipal archive, and which request data Elevate receives from MyDoc WIN once the integration method with AIRC is agreed.

Technical Risks

External Dependencies

The platform is meant to run at the municipality for years after this project, so we list the external parties it depends on in production, not the tools we use during development. We have no contract or SLA with any of them.

Dependency	**What we need from it**	**What can go wrong**	**Mitigation / alternative**
MyDoc WIN (AIRC), the CMA's document management system	Inspection requests registered by the CMA	AIRC releases updates on its own schedule that can change its data structures; no integration method confirmed	Documented export or API agreed with AIRC and the CMA, never its internal database; adapter with contract tests; file import (CSV/XML) as fallback
Authentik (identity)	Login and roles	Open-core vendor,	Only open-source features;

		frequent releases, paid enterprise features	standard OIDC/SAML so it can be replaced; pinned versions and tested upgrades
Autenticação.gov (ARTE, IP)	Owner login with Citizen Card or Digital Mobile Key;	Registration delays block go-live; an outage stops owners from logging in; Authentik may not support ARTE's SAML extensions;	Request registration early and develop against the pre-production environment; clear error page when the service is down; small adapter between Authentik and Autenticação.gov if needed
SeaweedFS (file storage)	Reports and documents	Some data-protection features are paid only	Backend uses only the S3 API, so the storage can be swapped; our own backups, no reliance on paid features
Hosting (CMA or municipal infrastructure)	95% availability	Environment changes outside our control	Containerized and provider-neutral (Docker Compose); documented requirements and upgrade procedure
Email delivery	Deadline alerts	Failed delivery causes missed legal deadlines	Standard SMTP (CMA relay first); delivery status tracked; in-app notices as a second channel
Open-source stack (React, Vite, TypeScript, Nginx, FastAPI, PostgreSQL, Docker)	Everything else	Abandoned libraries, breaking changes, vulnerabilities	Widely used versions, pinned and updated on a schedule, automated vulnerability checks

Design rule: no pay-per-use service on the critical path, so a price change by a third party cannot affect running costs or availability.

Integration with the CMA System

The platform must receive inspection requests from the municipality's existing system, MyDoc WIN, supplied by AIRC, without requiring changes to it. We have not yet confirmed how the data can be exchanged (API, export or file), and AIRC updates MyDoc on its own schedule, so an update could break an integration that depends on its internals.

To mitigate this, we will agree with AIRC and the CMA on a documented exchange mechanism and avoid reading MyDoc's database directly. The integration will sit behind an adapter in the backend with contract tests, so a change on the MyDoc side affects a single module. We will develop against a local MyDoc test instance, if AIRC's license allows it, and keep a file import as a fallback.

Technical Debt and Maintainability

Debt we are already taking on, and how we plan to repay it before the platform goes into production.

Decision	**Debt it creates**	**Refactoring plan**
Business rules (validity, status, deadlines) implemented in the	Rules end up duplicated and untested once a server exists; a change in the law means	Move the rules into one configurable module with unit tests; the server enforces them

prototype front end; periodicity depends on building type (2, 4 or 6 years) and is missing from the CMA's Excel	editing code in several places	and the front end only displays
Data model shaped by prototype sample data	Mismatch with the CMA's real Excel records, causing rework at migration	Derive the schema from anonymized real records; versioned database migrations from the first backend commit
Tests tied to sample data and fixed dates	Tests break whenever data changes, so people stop trusting them	Generate fixtures; test rules with relative dates; run tests in CI on every merge
Accessibility and usability fixes postponed	Known barriers from the accessibility matrix reach the MVP	Fixes part of the definition of done; accessibility check repeated before each release
Knowledge concentrated in the team	After the course nobody can maintain the system	README, architecture decision records and an operations runbook; handover session to the CMA
LLM-generated code	Inconsistent style and parts nobody fully understands	Review required before merge, linter and tests, no unreviewed generated code in core logic
Resilience delegated to the reverse proxy	Open-source Nginx only offers passive failure checks, so the MyDoc integration has no real circuit breaker	Timeouts, retries and a circuit breaker inside the adapter; update the architecture diagram
Five containers plus a database for about 100 requests per cycle	Upgrades, backups and monitoring fall on the municipal IT team	One Docker Compose file, runbook for upgrades and restores, health checks; revisit if the CMA cannot operate it

Brittle Components

The components that are most likely to fail, or to break when something around them changes.

Component	Why it is fragile	Impact of a failure	Mitigation
Integration with MyDoc WIN	Point-to-point link to a proprietary system we cannot change, updated by AIRC on its own schedule, with no confirmed API	New inspection requests stop reaching the platform, or arrive wrong, until someone notices	Abstraction: adapter behind one internal interface. Tests: contract tests against a local MyDoc test instance. Redundancy: manual entry and file import as fallback, plus monitoring that flags a silent integration
Excel import	Hand-maintained file with sheets going back to around 2000, no fixed format, and a scope column with fewer values than the request form	Duplicated, wrong or missing elevator records from the start, which conflicts with our requirement of zero duplicated or	Tests: import tested on anonymized real samples. Abstraction: validation and duplicate detection before saving. Redundancy: dry run with a report of rejected

		lost records	rows; the original file stays untouched
Deadline and notification engine	Inspection intervals depend on the building type (2, 4 or 6 years) and change after two periodic inspections; the CMA's Excel has none of the data needed to apply these rules, and the law can change	Wrong or missing alerts, so inspections are requested late or not at all and legal deadlines are missed	Abstraction: rules kept as configuration, and the next inspection date taken from the inspection certificate instead of calculated. Tests: unit tests for date calculation, including the change after two inspections. Redundancy: email plus in-app notices, with delivery status tracked

Licenses

Licenses of the components in the architecture. The open-source components use permissive licenses, compatible with an open-source project.

Component	**Role**	**License**
React	User interface	MIT
Vite	Front-end build tool	MIT
TypeScript	Front-end language	Apache-2.0
Nginx	Reverse proxy	BSD-2-Clause
FastAPI	Backend	MIT
PostgreSQL	Database	PostgreSQL License
Docker Engine	Containers	Apache-2.0
Authentik	Authentication	MIT (core). Enterprise features use a separate paid license, which we do not use

SeaweedFS	File storage	Apache-2.0. Some data-protection features are paid only
MyDoc WIN (AIRC)	CMA document management (external)	Proprietary. Use of a local test copy to be confirmed with AIRC

EXTRA

1. Integration with the CMA System

The platform must exchange inspection data with the municipality's existing system, provided by AIRC, without requiring changes to it. It is not yet clear how this integration will be done, whether through an API endpoint, file exchange or another mechanism.

To mitigate it, we will download the publicly available version of the AIRC system, run it in our own environment and develop the integration against it. Once the integration method and access is confirmed, we will only need to point the platform to the municipality's live instance, without rewriting the integration logic.

2. Data and Process Migration

The CMA currently keeps its elevator records in an Excel file, with sheets going back to around 2000, maintained by hand. Migrating this data may introduce duplicated, inconsistent or incomplete records, which conflicts with our requirement of zero duplicated or lost records.

To mitigate this, the platform will include an import feature for the CMA's Excel records, with validation and duplicate detection. The process migration will also be gradual: elevator owners and maintenance and inspection companies will be registered on the platform step by step, so the old and new processes can coexist during the transition.

3. Security and Personal Data Protection

The platform will store personal data (names, tax numbers and contacts of owners and condominiums) and sensitive operational data, such as building addresses and the results of failed inspections. A breach could expose this data, allow inspection results or deadlines to be tampered with, or take the platform offline, causing legal deadlines to be missed. It would also trigger GDPR obligations, including notifying the supervisory authority within 72 hours.

To mitigate this, access will be controlled by role and enforced on the server, not only in the interface. Data will be encrypted in transit and at rest, all changes to inspection records will be logged for audit, and user inputs and uploaded files will be validated. We will collect only the data the CMA needs, and use test data instead of real records in development and staging.

4. Data Loss and Backups

The platform will become the main register of the municipality's elevators and their inspection history. A hardware failure, a faulty update or an accidental deletion could cause records to be lost, which conflicts with our requirement of zero duplicated or lost records. Restoring the system after such an event could also take longer than the 95% availability target allows.

To mitigate this, we will run automatic backups of the database and the uploaded documents, store them outside the server that hosts the platform, and test the restore process regularly. A documented recovery plan will state who restores the system, in what order and within what time.

5. Licenses and Dependencies

The platform will be open source and built on third-party frameworks and libraries. If any of them uses a license that is not compatible with the project's license, or contains a known vulnerability, we may have to replace it late in the project.

To mitigate this, we will list the license of every framework and library in the LICENSE.md file of the repository and check compatibility before adopting each new dependency. Dependencies will be kept up to date, with automated checks for known vulnerabilities.

6. CMA Infrastructure Deploy

The prototype is planned to be deployed in a controlled environment at the municipality, but the CMA's infrastructure is still unknown. Incompatibilities could delay deployment or prevent the system from meeting its 95% availability target.

To reduce this risk, we will gather information about the CMA's infrastructure and its constraints in advance, package the whole system with Docker Compose, keep a staging environment that mirrors production, and deliver clear installation and operation documentation.

Risk	Probability	Impact	Priority
Integration with the CMA System	High	High	Critical
Data and Process Migration	High	Medium	High
CMA Infrastructure Deploy	Medium	High	High
Security and Personal Data Protection	Low	Critical	Critical
Data Loss and Backups	Low	High	Medium
Licenses and Dependencies	Low	Medium	Low